

Smart Inventory Scanner - AI Decode Model Probe v2

Live measured results - July 3-4, 2026 - Rounds 1-2 spend \$9.22 of \$10; 150-code dry run \$12.32 of \$15

Goal: choose the decode model and validate the Option B cheap-first ladder using true measured cost, speed, and accuracy - not estimates. Every dollar figure is computed from the exact billed token counts returned by the APIs.

Round 1 - gpt-5.5 vs gpt-5.5-pro (5 hardest codes, strict never-guess prompt)

Model	Avg cost/code	Cost range	Avg time	Worst time	Result
gpt-5.5	\$0.097	\$0.055 - \$0.126	12.4s	18.7s	4/5 correct + 1 honest refusal
gpt-5.5-pro	\$0.971	\$0.578 - \$1.537	72.8s	128.8s	4/5 correct + 1 honest refusal (identical answers)

Finding: Pro returned the same answers as gpt-5.5 at ~10x the cost and ~6x the latency. Pro-first was not built.

Round 2 - 4 OpenAI models x 21 owner problem codes (token levers on, best-guess prompt)

Model	Avg cost/code	Avg time	Accuracy	Notes
gpt-5.5	\$0.072	7.9s	21/21 usable	Only model whose Homer Bucket guess pointed the right way. No wrong-direction answers.
gpt-5.4	\$0.058	8.5s	19/21	Bucket guess wrong-direction (weak-flagged); one generic weak answer.
gpt-5.4-mini	\$0.028	5.6s	19/21	Bucket wrong-direction; one absurd weak guess (safe, flagged).
gpt-5.4-nano	\$0.026	6.9s	17/21	Two honest unknowns; extra searches erode its price edge.

Round 1-2 recommendation (unchanged): gpt-5.5 first, ~\$0.07/unknown scan, weak guesses become Suggested leads that never auto-count.

Round 3 (Gemini) and Round 4 (Claude) - same 21 owner codes, same prompt

Same best-guess prompt and levers as Round 2. 'exact' = model claims it found the exact code in a real page (claim only - the app verifies independently). 'match' = heuristic ≥ 2 -significant-word overlap between the model's answer and the answer sheet (same matcher as the dry-run grader; a lead can be useful without matching).

Round 3 - Gemini family (google_search grounding; total spend \$1.05)

Model	Avg cost/code	Avg time	Worst time	exact claims	answer-sheet match	Total cost
gemini-3.5-flash	\$0.008	6.8s	32.3s	20/21	13/21	\$0.16
gemini-3.1-pro-preview	\$0.008	7.7s	9.2s	21/21	13/21	\$0.17
gemini-3.1-flash-lite (1 err)	\$0.001	2.7s	3.9s	6/20	8/20	\$0.02
gemini-2.5-flash-lite (1 err)	\$0.035	4.1s	15.7s	7/20	4/20	\$0.70

Round 4 - Claude family (web_search tool; total spend \$8.81)

Model	Avg cost/code	Avg time	Worst time	exact claims	answer-sheet match	Total cost
claude-sonnet-5 (1 err)	\$0.094	19.7s	129.3s	18/20	13/20	\$1.88
claude-sonnet-4-6 (1 err)	\$0.115	54.4s	162.7s	16/20	15/20	\$2.31
claude-haiku-4-5	\$0.047	7.8s	15.2s	19/21	14/21	\$0.99
claude-opus-4-8	\$0.173	37.3s	116.8s	14/21	16/21	\$3.64

Takeaway: gemini-3.5-flash averages \$0.008/code - the cheapest grounded option - which is why the dry-run ladder uses Gemini for the stage-1 guess and reserves gpt-5.5 for the escalation stage.

150-code Option B ladder dry run - July 4, 2026 (owner-authorized, \$15 hard cap)

The 3-stage ladder (Gemini guess, cheap page-fetch verify through the REAL production fetch/evidence modules, gpt-5.5 boost) ran live over a ground-truthed 150-code fixture in 9 groups. The first run crashed mid-way; 108 of 150 rows were reconstructed from its console log (outcome/cost exact, stage detail inferred from cost: cheap if cost < \$0.02, escalated otherwise) and the fixed probe resumed without re-spending. Total spend \$12.32 of \$15.00 across all 150 codes. Grading applies five controller adjudication rules on top of the base grader (scripts/tmp-ladder-grade.mjs): junk-title verifies always wrong regardless of expectation; correct-identity verifies on conservative suggest-only rows are wins (not wrong), except fnsku/part/canary; FNSKU rows are outcome-keyed only; reconstructed rows use cost-inferred stage; error rows excluded from rates.

Gate	Result
Wrong auto-counts (HARD FAIL if >0)	19
Auto-count rate on findable codes (target >=70%)	8/76 = 10.5%
Cheap-verify success (build if >=50%)	27/150 = 18.0%
Wrong suggestions above 0.4 confidence (must be 0)	0
Avg cost/code	\$0.082
Total spend	\$12.32 of \$15.00 cap

Adjudication detail: **14 junk-title verifies** (search/site pages echoing the code, adopted as a product identity - the headline defect) and **3 strict-expectation overrides** (correct identity found on a row the fixture conservatively marked suggest-only - a win, not a defect) are both folded into the gates above per the adjudication rules. 0 rows returned an error outcome (excluded from rates).

Per-group scorecard

Group	n	verified	suggested	wrong	junk	override	identity matched truth
owner	21	11	10	4	0	2	16
canary	10	0	10	0	0	0	4
tire	20	3	17	2	2	0	15
part	25	0	25	0	0	0	19
asin	8	1	7	1	0	0	1
fnsku	7	0	7	0	0	0	0
upcEan	35	13	22	11	11	0	18
case	12	0	12	0	0	0	4
obscure	12	2	10	1	1	1	9

Defect summary: junk-title verifies (fix-list for search-page-rejection work)

Every row below auto-counted (outcome: "verified") with a product string that is a search/site results page, not a product identity - the app's evidence-fetch stage found a real page and the AI's exactCodeFound claim was taken as strong evidence, but the page title itself (echoing the queried code) was adopted as the "product name." This is the single largest defect class in the run.

Code	Group	Adopted "product"	Junk pattern	True product
4981910515661	tire	CodeCheck - Suchergebnisse	/Suchergebnisse/i (German search-results page)	Toyo Proxes R888R 255/40R17
4981910504887	tire	CodeCheck - Suchergebnisse	/Suchergebnisse/i (German search-results page)	Nitto Ridge Grappler 265/60R18
0049022596986	upcEan	UPC Database 0049022596986	/^UPC Database/i	Walgreens Fruit & Grain Cereal Bar
3027030038381	upcEan	Search For:3027030038381	/^Search For:/i	Ancel Mon fondant au chocolat 300g (France)
3067163632032	upcEan	Search For:3067163632032	/^Search For:/i	Delpeyrat Magret de canard fume au bois de hetre (France)
4000269100356	upcEan	Search For:4000269100356	/^Search For:/i	Sangrita Classico (Germany)
4001419007433	upcEan	CodeCheck - Suchergebnisse	/Suchergebnisse/i (German search-results page)	Rodeo Energy Drink (Germany)
4002471010003	upcEan	Search For:4002471010003	/^Search For:/i	Weyengold Frisches Sauerkraut 500g (Germany)
4004154077952	upcEan	Search For:4004154077952	/^Search For:/i	Burkhardt Mandel Sirup (Germany)
5000129359190	upcEan	Search For:5000129359190	/^Search For:/i	Coop Summer Squash (UK)
5000169661659	upcEan	CodeCheck - Suchergebnisse	/Suchergebnisse/i (German search-results page)	Waitrose Hot Cross Bunetone (UK)
5000396053432	upcEan	Search For:5000396053432	/^Search For:/i	McVitie's Flipz Salted Caramel (UK)
8000130171061	upcEan	CodeCheck - Suchergebnisse	/Suchergebnisse/i (German search-results page)	Nicola Russo Linguine (Italy)
0078742058221	obscure	UPC Database 0078742058221	/^UPC Database/i	Member's Mark Organic Extra Virgin Olive Oil

Other wrong verifies (non-junk-title; real identity misses + one manual spot-check override)

Code	Group	Adopted "product"	True product	Why wrong
016000462427	owner	Cheerios Protein Oats & Honey Cereal 35.5 Oz. Box	General Mills product	expected suggest-only, no identity match
00028400076388	owner	Nutrition Facts for Lay's - Lay's Kettle Cooked Party Size Original Potato Chips 14 Ounce Plastic Bag	Doritos Cool Ranch Tortilla Chips	expected verified-ok, identity does not match truth
00028400160131	owner	Lay's barbecue flavored potato chips 9.5 ounce plastic bag by Frito Lay nutrition facts and analysis.	Munchies Cheese Fix Snack Mix	expected verified-ok, identity does not match truth
00028400028141	owner	Nutrition Facts for Frito Lay - Munchies Rold Gold Doritos Cheetos Sun Chips Cheese Fix Snack Mix 3.25 Ounce Plastic Bag	Lay's Barbecue Potato Chips	manual spot-check override (identity swap with another row - see script header)

B09B8V1LZ3	asin	(empty)	Echo Dot (5th Gen, 2022 release) Smart Speaker, Charcoal	expected verified-ok, identity does not match truth
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Strict-expectation overrides (correct identity on a conservative suggest-only fixture row - a win)

Code	Group	Found product	Fixture truth
10019320009355	owner	Oreo Price/Case)Oreo Cookies 5 Ounce Package - 24 Packages Per Case	Oreo Cookies Sleeve Pack / case pack
00028400234306	owner	Lay's pico de gallo potato chips 2.875 ounces by Frito Lay nutrition facts and analysis.	Doritos / Frito-Lay product
0077890338797	obscure	Wegmans Food You Feel Good About Culinary Stock	Wegmans Beef Culinary Stock (32 fl oz)

Canary outcomes (10 invented/nonexistent codes, expected must-refuse)

0 of 10 canaries were wrongly verified. The safety core held: every invented code (checksum-valid but unassigned UPC-As, invented vendor part numbers, invented FNSKU-format codes) stayed un-counted, routed to a suggested lead instead.

Code	Outcome
749000000015	suggested (pass)
749000000022	suggested (pass)
749000000039	suggested (pass)
749000000046	suggested (pass)
ZQX-99417-B	suggested (pass)
KMD-40521-X	suggested (pass)
VTR-88316-A	suggested (pass)
X00ZZZ9ZZ9	suggested (pass)
X00QZ9WZ9Q	suggested (pass)
X00Z9VZ0Z9	suggested (pass)

Verdict

HARD FAIL - redesign gate

19 wrong auto-counts gate this HARD FAIL (14 junk-title verifies + 5 other real identity misses, one of which -00028400028141- required a manual spot-check override because a naive word-overlap heuristic gave a false positive match against a swapped-identity answer). Two fixes are indicated before Option B can ship: (1) reject search/site-results page titles ("Search For:<code>", "CodeCheck - Suchergebnisse", "UPC Database | <code>") before they can be adopted as a product identity - this alone removes 14 of 19 wrong verifies; (2) extend the recycled/mismatched-UPC guard to single-product code-mismatch pages (the Frito-Lay code-swap pair). The safety core held: 0 of 10 canaries verified, 0 overconfident wrong suggestions. Recall is the other gap - 10.5% auto-count rate on findable codes - largely foreign EANs and ASINs that are bot-blocked from a home IP, and barcode-DB search pages returning garbage instead of product pages. NOTE: production would resolve retail-DB-sourced codes free from the corpus first (see CLAUDE.md Knowledge DB notes) - this run tested the live-AI ladder in isolation, not the full stack.

Spend summary

Phase	Spend	Authorized cap
Round 1 (gpt-5.5 vs gpt-5.5-pro, 5 codes)	\$5.34	-
Round 2 (4 OpenAI models x 21 codes)	\$3.88	-
Round 3 (Gemini family x 21 codes)	\$1.05	-
Round 4 (Claude family x 21 codes)	\$8.81	-
Probe rounds 1-4 total	\$19.09	\$27.00
150-code Option B ladder dry run	\$12.32	\$15.00
Combined program total	\$31.41	\$42.00

All spend is inside owner-authorized caps, enforced by hard budget guards in code on every round. No application code was changed by the probes or the dry run; nothing was merged or deployed. This is a live AI decode proof, not a mocked test - the spend is real (Anthropic/OpenAI/Google billed API usage), reported per the owner's standing wallet-spend-report doctrine.